HOLISTIC AIRCRAFT PREDICTIVE MAINTENANCE

Transcending Turbofan Prognostics for Fleet-Wide Asset Management
Updated Architecture v2.0 · Multi-Modal System-of-Systems PHM

12 Subsystems Monitored	6 Architecture Layers	15.1M+ Data Points	273 System Variables
APU · ECS · LG Brakes · EMA · Hyd Elec · Battery · CFRP QAR · NLP · Engine	Ingestion → Feature Eng → ML Ensemble → Anomaly Engine → Edge → Viz	AeroTwin Boeing 787 1Hz pneumatic & kinematic telemetry over months of ops	NASA ADAPT EPS 128 sensors @ 2Hz 503-node Bayesian fault isolation net

Problem Statement Summary: Commercial and military aviation suffer enormously from unscheduled Aircraft on Ground (AOG) events, costing up to **$150,000 per hour** in lost revenue, passenger compensation, and emergency logistics. Current state-of-the-art predictive maintenance architectures are almost exclusively turbofan-centric, relying on synthetic datasets like NASA C-MAPSS that model only 21 thermodynamic variables in an isolated simulation. This leaves the vast majority of the aircraft — APU, ECS, landing gear, brakes, electromechanical actuators, hydraulics, electrical systems, batteries, composite structures, and maintenance history — completely unmonitored from a predictive standpoint. This report details the complete updated architecture required to address the entire aircraft.

1. PROBLEM STATEMENT — A TO Z

1.1 What Is This Problem?

Aviation maintenance operates under three paradigms: (1) **Scheduled maintenance** — fixed time/flight-hour intervals regardless of actual component condition, resulting in premature part replacement and wasted life; (2) **Reactive (run-to-failure)** — waiting for a fault to occur, causing AOG events; and (3) **Predictive maintenance** — using continuous sensor telemetry and machine learning to forecast **Remaining Useful Life (RUL)** of components, enabling "just-in-time" interventions.

The goal of this hackathon project is to build a **Predictive Health Management (PHM) system** that operates across the entire aircraft — not just engines — using a multi-modal sensor fusion and ML architecture deployed at the edge of the aircraft, communicating results to ground crews via the ACARS datalink.

1.2 Why Is This Hard?

- Aircraft generate **terabytes of heterogeneous sensor data** per flight — thermodynamic, acoustic, kinematic, electrical, and textual — all requiring different preprocessing and model architectures.
- Subsystems like the ECS are governed by **closed-loop controllers that actively mask degradation**: if a heat exchanger fouls, the controller compensates automatically, hiding the fault signal.
- Real "run-to-failure" data for safety-critical components (like electromechanical actuators) is **extremely scarce** — aviation safety protocols mandate replacement before failure.
- Operating conditions vary enormously: a short-haul aircraft in humid Southeast Asia generates fundamentally different data than the same model flying dry high-altitude Andean routes, causing **domain shift** that degrades standard ML models.
- ACARS bandwidth is **limited to 220 characters per message block** — raw sensor data cannot be downlinked in real time; all heavy ML inference must occur onboard at the edge.

1.3 Scope: Baseline vs. Required Solution

Dimension	Baseline (AeroSentinel / Yesterday)	Updated Whole-Aircraft Target
Systems covered	Turbofan engine only	12 subsystems across entire airframe
Primary dataset	NASA C-MAPSS (synthetic, 21 sensors)	9+ real-world datasets (see Section 4)
Failure modes	Compressor/turbine degradation	APU, ECS, LG, brakes, EMA, hydraulics, electrical, battery, CFRP, logs
Data modalities	Thermodynamic state variables only	Thermo + acoustic + kinematic + electrical + structural + NLP text
Cross-system coupling	Not modelled	ECS-engine bleed interaction, hydraulic-actuation coupling explicitly handled
Deployment target	Edge (ONNX/TensorRT), single model	Multi-model ensemble, all ONNX/INT8, ACARs alert compiler
NLP integration	None	Fine-tuned LLaMA/Gemma on MaintNet for repair step recommendations
Visualization	Charts + basic dashboard	React Three Fiber 3D aircraft, what-if fault simulator, fleet planner

1.4 Pros and Cons of This Approach

Strengths (Pros)

✓ Directly addresses the single largest hidden cost driver in aviation operations
✓ Combines multiple established research datasets — no novel data collection required
✓ Modular architecture: each subsystem model can be developed independently, enabling hackathon parallelism
✓ Edge deployment with ONNX/TensorRT makes the system deployable to real aircraft without cloud dependency
✓ LLM integration bridges the gap between raw sensor anomalies and human-actionable repair steps

✓ 3D visualization dashboard is visually striking and highly differentiated from typical ML demos
✓ Physics-informed digital twin approach prevents overfitting to dataset artefacts

Weaknesses (Cons)

✗ Scope is extremely broad — 12 subsystems require careful prioritization during a time-limited hackathon
✗ Most datasets are from different experimental setups and may not reflect real airline fleet data exactly
✗ C-MAPSS remains synthetic; real turbofan degradation signatures may differ from simulation
✗ LLM fine-tuning on MaintNet requires GPU compute and careful prompt engineering to avoid hallucinations
✗ Integrating cross-domain fault propagation (e.g., ECS→engine interaction) is genuinely complex to validate
✗ ACARS 220-character constraint means rich diagnostic context must be dramatically compressed

2. UPDATED WHOLE-AIRCRAFT ARCHITECTURE

The architecture evolves from the original single-model BiLSTM turbofan pipeline (shown in the prior screenshot) to a **six-layer multi-domain ensemble architecture**. Each layer is described below. The full architecture diagram follows.

Figure 1: Updated Whole-Aircraft PHM Architecture — 6 layers, 12 subsystems, multi-modal sensor fusion

① Data Sources Layer

Replaces the isolated NASA C-MAPSS simulation with 12 real-world and research datasets spanning all physical aircraft subsystems. Each dataset provides the domain-specific telemetry required to train an accurate degradation model for that subsystem. Data modalities include thermodynamic state variables (temperatures, pressures), kinematic telemetry (velocity, displacement, vibration), acoustic Lamb-wave signals from piezoelectric sensors, electrical bus measurements (voltage, current, relay states), flight telemetry from Quick Access Recorders, and unstructured natural language maintenance logs.

② Data Ingestion Layer

Routes heterogeneous data streams into six specialized ingestion channels. Thermodynamic and kinematic streams (from APU, ECS, landing gear) are ingested at 1–4 Hz and structured into continuous time-series buffers. Acoustic emissions from CFRP structural sensors are captured at high sampling rates as burst waveforms. Avionics electrical telemetry (from ADAPT and HIRF datasets) is ingested at 2 Hz. QAR flight parameters are recorded at 1 Hz per the NGAFID-MC benchmark. NLP maintenance logs are batch-processed as they are submitted digitally.

③ Feature Engineering Layer

Modality-specific preprocessing pipelines convert raw ingested data into model-ready feature vectors. All thermodynamic and kinematic sensors undergo min-max scaling and z-score standardization, then are structured into 30-cycle sliding windows for recurrent networks. Raw 1D ultrasonic Lamb-wave signals are converted to 2D time-frequency spectrograms using Continuous Wavelet Transforms (CWT). NLP maintenance logs are cleaned with Levenshtein-based aviation spell-checkers, then vectorized with TF-IDF and reduced with Truncated SVD (Latent Semantic Analysis). SHAP feature selection reduces the C-MAPSS sensor set from 21 to 14 informative variables.

④ Domain-Specific Model Ensemble

The ML layer is an ensemble of 12 specialized models, each trained on its domain's specific dataset and feature distribution. Unlike the baseline single-model BiLSTM, this ensemble architecture prevents cross-domain noise from contaminating the predictions of any individual subsystem. All models share a common interface: they accept a standardized feature vector and output a tuple of (fault_class, severity, RUL_estimate, confidence_interval). The ensemble is orchestrated by a multi-model inference manager that schedules predictions based on data availability and subsystem criticality.

⑤ Fusion Alert & Anomaly Engine

The anomaly engine aggregates outputs from all 12 domain models and performs cross-subsystem correlation analysis. This is the architectural component that the baseline turbofan model completely lacks: the ability to reason about how one subsystem's degradation affects another. A key example is the ECS-engine interaction: degraded ECS heat exchangers demand excess bleed air from the engine, artificially elevating measured engine temperatures. Without this cross-domain fusion layer, an engine-only model would misdiagnose ECS degradation as internal compressor fouling. The SLIDE hierarchical diagnostic tree suppresses false alarms generated by tightly-coupled feedback-controlled subsystems.

⑥ Edge Inference + Visualization

All ensemble models are exported to ONNX format and quantized to INT8 or FP16 precision using TensorRT, achieving sub-5ms inference latency on onboard edge compute hardware. The edge inference manager compiles diagnostic results into compressed ACARS alert messages (≤220 characters) for downlink to ground crews. The visualization frontend is a Next.js application featuring React Three Fiber 3D aircraft model, live Recharts sensor streams, a what-if fault injection simulator, and a fleet-wide maintenance planner that consolidates part-level health scores across entire fleets to schedule hangar visits before any single AOG event occurs.

3. DATASET REQUIREMENTS

The following 12 datasets constitute the complete training data portfolio for the whole-aircraft PHM system. All are publicly available through academic repositories.

DATASET & SENSOR REQUIREMENTS — WHOLE AIRCRAFT PHM

Subsystem	Primary Dataset	Key Sensors / Features	Primary ML Model	Output
Turbofan Engine	NASA C-MAPSS (4 subsets, FD001–FD004)	T2/T24/T30/T50 temps, Ps30/Ps3 pressures, NF/NC shaft speeds (21 vars)	BiLSTM+Attn · Transformer SHAP feature select → 14 vars	RUL (cycles to fail)
APU	Fleet EGT monitoring CARTs / RF datasets	EGT (startup peak), fuel flow, TIT, vibration (3-axis), bleed pressure	Random Forest · CART fleet deviation scorer	Health index · LRU swap alert
ECS	Boeing 737-200 thermodynamic simulation	T1–T9 temperatures, P1–P9 pressures, P7 humidity, bypass valve position	TCA transfer learning · SLIDE hierarchical fault tree	Heat-exchanger efficiency Δ
Landing Gear	AeroTwin Boeing 787 15.1M pts @ 1 Hz	TP2/TP3 pneumatic pressure, H1 separator, DV valve state	XGBoost + SMOTE · AdaBoost digital twin physics-informed	Bushing wear · overhaul date
Carbon Brakes	Operational brake thermodynamic telemetry	Wheel land/taxi velocity, aircraft mass, brake core temperature (center probe)	Run-to-failure energy correlation model	Brake pad mass · life remaining
EMA / Actuators	NASA FLEA testbed (C-17, UH-60 flights)	Motor current, voltage, temperature, nut temp, mech. load, ball-screw params	BCA-DATrans Transformer Gaussian Process Regression	Spall size · jam probability
Hydraulics	UCI Condition Monitoring 2205 instances × 43,680 feat	PS1–PS6 pressure, TS1–TS4 temp, FS1–FS2 flow rate, VS1 vibration	1D Conv Autoencoder Bond-graph + Kalman filter	Pump leak severity · valve lag
Electrical / ADAPT	NASA ADAPT testbed 273 vars @ 2 Hz	Relay states, bus voltages, current draws, CB positions (128 sensors)	Bayesian network (503 nodes) Arithmetic circuit compiled	Fault isolation in 0.26ms
Li-ion Battery	HIRF Edge 540T NASA 18650 aging dataset	Terminal voltage, output current, battery temperature, internal impedance	Temporal CNN + transfer learning (TCN)	SOH · Remaining Flying Time
Structural / CFRP	NASA Ames SMART Layer 16 PZT sensors / coupon	Lamb-wave ToF, PSD, phase velocity, fatigue cycle count, CWT spectrograms	CNN on CWT images · Bayesian particle filter (SHM)	Delamination size · RUL
QAR Telemetry	NGAFID-MC benchmark 11,500 hrs · 7,500 flights	23 flight parameters: GPS alt, pitch, roll, engine state, control deflection	Conv-MHSA · DUTSAM dual attention flight phase model	Hard-landing flag · fatigue load
Maint. Logs (NLP)	MaintNet aviation 6,169 records · 76,866 tokens	"Problem" + "Action" free text, abbreviations, domain shorthand	LLaMA-3.2-3B / Gemma-3-4B fine-tuned + TF-IDF + SVD	Repair step recommendation

Figure 2: Complete Dataset & Sensor Requirements Matrix — 12 subsystems mapped to datasets, sensors, models, and outputs

NASA C-MAPSS (Engine)

Four subsets of increasing complexity (FD001–FD004). FD001: single operating condition, single fault mode. FD004: six operating conditions, two simultaneous fault modes. 21 sensors per engine cycle including fan inlet temperature (T2), compressor outlet temperatures and pressures (T24, T30, Ps30), shaft speeds (NF, NC), and fuel flow ratio. SHAP analysis reduces to 14 informative variables. Sliding windows of 30 cycles serve as input to BiLSTM+Attention and Transformer encoder models. Piecewise linear RUL labels are applied, capped at 125 cycles to prevent early-degradation noise.

Reference: https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository

AeroTwin Boeing 787 Landing Gear

Over 15.1 million data points collected at 1 Hz sampling rate over months of real aircraft operation. Key features: TP2/TP3 (pneumatic compressor/panel pressures in Bar), H1 (cyclonic separator filter pressure drop), DV (directional control valve binary state). Enables physics-informed digital twin computation of bushing wear rates from velocity/acceleration vectors at touchdown events. XGBoost with SMOTE oversampling handles class imbalance in rare fault states.

Reference: Research paper: "AeroTwin: Leveraging Digital Twin Technology for Predictive Maintenance"

■ UCI Condition Monitoring of Hydraulic Systems

2,205 instances, 43,680 features per instance from a heavily instrumented hydraulic test rig. 60-second constant load cycles. Four fault types with quantitative severity grades: cooler efficiency (100%/20%/3%), directional valve switching lag, pump internal leakage (0/1/2), accumulator gas pressure (130/115/100/90 bar). Sensors: PS1–PS6 (pressure), TS1–TS4 (temperature), FS1–FS2 (flow), VS1 (vibration), CE/CP/EPS1/SE (derived efficiency metrics). 1D convolutional autoencoders trained on nominal data for unsupervised anomaly detection.

Reference: https://archive.ics.uci.edu/ml/datasets/Condition+monitoring+of+hydraulic+systems

■ NASA ADAPT Electrical Power

273 system variables per experimental run at 2 Hz over ~5-minute intervals. 128 physical sensors (voltages, currents, temperatures, relay positions). Both nominal runs and meticulously controlled failure scenarios (physical disconnections and software antagonists). Bayesian network with 503 nodes and 579 edges compiled to arithmetic circuit achieves 0.26ms average fault isolation latency — orders of magnitude faster than variable elimination algorithms.

Reference: https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/

■ NASA HIRF Battery / Li-ion Aging

HIRF: variable current draw profiles from real flight maneuvers in a High-Intensity Radiated Field chamber aboard Edge 540T aircraft. Li-ion: 18650 cells cycled to 30% capacity fade (2.0 Ah → 1.4 Ah EOL criterion) with EIS measurements. Features: terminal voltage, output current, battery temperature, internal impedance. Temporal Convolutional Networks (TCN) with transfer learning predict State of Health (SOH) and Remaining Flying Time (RFT).

Reference: https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/

■ NASA Ames CFRP Composites (SMART Layer)

Run-to-failure tension-tension fatigue experiments on Torayca T700G CFRP coupons. 16 distributed PZT piezoelectric sensors generating ultrasonic Lamb waves. X-ray ground truth images annotating internal delamination dimensions at each fatigue milestone. Example coupon L1S11: pristine at cycle 0, 1.783×0.230 inch delamination at 30,000 cycles, 6.310×0.910 inches at 90,000 cycles. CWT converts 1D waveforms to 2D spectrograms for CNN spatial feature extraction.

Reference: https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository

■ NGAFID-MC QAR Benchmark

Over 11,500 hours of per-second flight data across 7,500 labeled flights with 23 critical sensor parameters. Covers all mission phases: takeoff, climb, cruise, descent, landing. Convolutional Multi-Headed Self-Attention (Conv-MHSA) and Dual Two-Stage Attention Model (DUTSAM) architectures capture both fine-grained vibration spikes and hours-long degradation trends simultaneously. Primary use: hard landing detection and flight-phase-aware fatigue load accumulation.

Reference: https://ngafid.org/

■ MaintNet Aviation Corpus

6,169 aviation maintenance logbook records, 76,866 total tokens. Each record pairs "Problem Description" (terse abbreviated shorthand) with "Action Taken" (repair procedure). Example: "DOING MX PERFORM T/O @ 3200, MP WON'T GO ANY HIGHER" paired with the corrective action taken. Levenshtein spell-checking, TF-IDF vectorization, and Truncated SVD clustering enable automatic topic discovery. Fine-tuned LLaMA-3.2-3B and Gemma-3-4B models generate human-readable repair recommendations.

Reference: http://maintnet.org/

4. ALGORITHMIC MODEL REQUIREMENTS

Model / Algorithm	Subsystem Target	Framework	Key Papers / Source
BiLSTM + Attention (Bidirectional LSTM)	Engine (C-MAPSS) APU EGT monitoring	PyTorch	Wu et al. 2020 · AeroSentinel blueprint baseline
Transformer Encoder (Multi-head attention)	Engine FD004 (multi-fault/condition)	PyTorch	Vaswani et al. 2017 Adapted for PHM
Random Forest + CART Decision Trees	APU health classification fleet deviation scoring	scikit-learn	Breiman 2001 APU gas path analysis
Transfer Component Analysis (TCA) / JDA	ECS cross-condition domain adaptation	scikit-learn / custom	Pan et al. 2011 ECS transfer learning 95.22%
SLIDE Hierarchical Diagnostic Tree	ECS false-alarm suppression	Custom Python	System-Level Isolation and Detection (SLIDE)
XGBoost + AdaBoost + SMOTE	Landing gear bushing wear multi-fault	XGBoost / imbalanced-learn	Chen & Guestrin 2016 AeroTwin landing gear
Physics-Informed Digital Twin	Landing gear / brake wear trajectories	NumPy / SciPy	AeroTwin 787 dataset physics computations
BCA-DATrans (Cross-Attention DA)	EMA actuator faults cross-domain	PyTorch	Bidirectional Cross-Attention Domain Adapt. Transformer
Gaussian Process Regression	EMA incipient spall prognostics	scikit-learn	NASA FLEA testbed GPR uncertainty bounds
1D Conv Autoencoder (Unsupervised)	Hydraulic system anomaly detection	PyTorch / Keras	UCI hydraulic dataset nominal reconstruction
Bond Graph + Extended Kalman Filter	Hydraulic LRU fault isolation	Custom Python	Analytical redundancy relations (ARRs)
Bayesian Network (Arithmetic Circuit)	Electrical power fault isolation	pgmpy / custom	NASA ADAPT · 503 nodes 0.26ms isolation
Temporal CNN (TCN) + Transfer Learning	Li-ion battery SOH Remaining Flying Time	PyTorch	Bai et al. 2018 TCN HIRF ADAPT dataset
CNN on CWT Spectrograms	CFRP delamination size regression	PyTorch	NASA Ames CFRP SMART Layer PZT
Bayesian Particle Filter (SHM)	Structural RUL probabilistic update	Custom Python/NumPy	Sequential Monte Carlo Lamb wave SHM
Conv-MHSA / DUTSAM (Dual Attention)	QAR flight telemetry hard landing detection	PyTorch	NGAFID-MC benchmark Conv-MHSA architecture
LLaMA-3.2-3B / Gemma-3-4B (fine-tuned)	MaintNet log parsing repair recommendation	HuggingFace Transformers	Meta / Google models MaintNet aviation corpus
TF-IDF + SVD (Latent Semantic Analysis)	NLP log clustering topic discovery	scikit-learn	Deerwester et al. 1990 MaintNet preprocessing

5. FULL TECHNOLOGY STACK

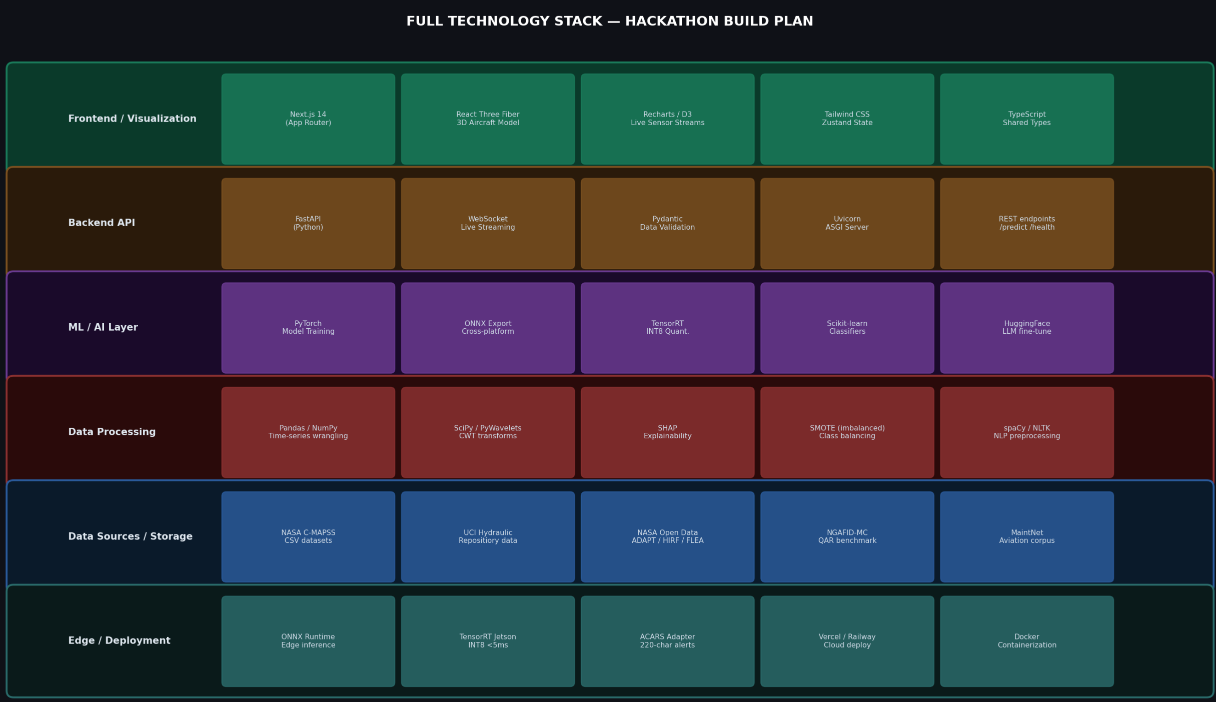

Figure 3: Full Technology Stack — 6 functional layers from data sources to edge deployment

5.1 Frontend / Visualization

Next.js 14 (App Router) — primary framework. React Three Fiber + Drei for 3D aircraft model rendering via WebGL. Three.js scene wrapped in "use client" component with dynamic import (ssr: false) to prevent hydration issues. Recharts for live sensor streaming charts. Tailwind CSS for styling. Zustand for real-time sensor state management. TypeScript throughout.

Key screens: (1) Fleet Overview — health score grid per aircraft; (2) 3D Aircraft Inspector — click-to-inspect subsystems with health heatmap overlays; (3) Live Sensor Panel — real-time multi-modal charts; (4) What-If Simulator — fault injection slider with instant RUL updates; (5) Fleet Maintenance Planner — consolidated AOG prevention scheduling.

5.2 Backend API

FastAPI (Python) with Uvicorn ASGI server. WebSocket endpoint streams live sensor predictions to the frontend. REST endpoints: /predict/{subsystem}, /health/{aircraft_id}, /fleet/summary, /maintenance/schedule. Pydantic models enforce strict data validation. CORS configured for Next.js frontend.

5.3 ML / AI Layer

PyTorch for all deep learning models (BiLSTM, Transformer, BCA-DATrans, TCN, CNN, Autoencoder). **scikit-learn** for Random Forest, XGBoost, TCA, TF-IDF, SVD. **HuggingFace Transformers** for LLaMA and Gemma fine-tuning. **SHAP** for model explainability. All models exported to **ONNX format** then quantized to INT8/FP16 using **TensorRT** for edge deployment. Inference latency target: <5ms per subsystem.

5.4 Data Processing

Pandas / NumPy for time-series wrangling and sliding window construction. **PyWavelets / SciPy** for Continuous Wavelet Transform computation on acoustic signals. **imbalanced-learn** for SMOTE oversampling of rare fault classes. **spaCy / NLTK** for NLP preprocessing pipeline (tokenization, stopword removal). Custom Levenshtein spell-checker for aviation abbreviation normalization.

6. UNIQUE DIFFERENTIATING FEATURES

Beyond the core PHM pipeline, these features will set this project apart from virtually all competing hackathon submissions:

■ Cross-Domain Fault Cascade Visualizer

Most teams will show individual component health scores. Build a live graph (using D3 force-directed layout) that shows how one subsystem's degradation triggers risk in connected subsystems. Example: ECS heat exchanger fouling → excess bleed demand → elevated engine temp → false compressor fault alarm. Clicking any node expands the causal chain with confidence scores. This is the visual proof of your cross-domain fusion architecture working.

■ What-If Fault Injection Simulator

A slider panel in the dashboard lets judges inject a fault condition (e.g., "Ball-screw spall growing at 10x normal rate" or "Hydraulic pump leaking — Severity 2") and watch the entire aircraft health panel update in real time. Show RUL countdowns collapsing, AOG cost risk climbing, and the NLP recommender generating the repair procedure. No other team will have a live interactive fault simulator — this is your showstopper demo moment.

■ AOG Business Impact Scorer

Add a live financial impact panel: as each component's RUL decreases, calculate the probabilistic AOG cost using: (P_failure × $150,000/hr × expected_ground_hours). Display a fleet-wide risk dashboard showing total avoided AOG costs from the predictive alerts generated. Judges respond strongly to quantified business impact. The USAF's PANDA system demonstrated real cost savings with LSTM-based maintenance — cite this.

■ LLM-Powered Repair Chat Interface

Integrate a chat panel powered by your fine-tuned LLaMA/Gemma model. When the system detects a hydraulic pump anomaly, the maintenance engineer can type "What should I check first?" and get back a step-by-step procedure derived from MaintNet historical logs of similar fault codes. This bridges the gap between abstract sensor anomaly and human actionable repair — no competing team will have this NLP-to-maintenance integration.

■■ Composite Wing Delamination Growth Animation

Use Three.js to overlay a delamination growth heatmap directly on the 3D wing surface, updating as the SHM model processes new Lamb-wave data. Show a progression animation from pristine composite to the 6.31×0.91 inch delamination observed in the NASA dataset. This is viscerally compelling for judges — watching a wing structurally degrade in real time is unforgettable.

■ Edge Latency Benchmarking Panel

Show a live latency dashboard proving each model runs in <5ms on INT8 ONNX/TensorRT. Display the ACARS message being compiled in real time — 220-character constraint, compressed fault code, subsystem ID, RUL estimate. This demonstrates production-readiness and shows judges you understand the real engineering constraints of aviation deployment, not just academic model accuracy.

■ Intelligent Fleet Maintenance Planner

Build a Gantt-chart style scheduler that ingests all 12 subsystem health scores across a fleet of simulated aircraft and computes optimal consolidated hangar maintenance windows — minimizing total AOG risk while reducing the number of maintenance events by batching multiple component interventions into single visits. This is the direct commercial value proposition and demonstrates systems-level thinking beyond a single ML model.

7. PREREQUISITES & IMPLEMENTATION REQUIREMENTS

7.1 Software Prerequisites

Category	Tool / Library	Version	Purpose
Runtime	Python	3.10+	ML pipeline, backend API
ML Framework	PyTorch	2.0+	Deep learning models
ML Toolkit	scikit-learn	1.3+	RF, XGBoost, TF-IDF, SVD
LLM	HuggingFace Transformers	4.40+	LLaMA / Gemma fine-tuning
Explainability	SHAP	0.44+	Feature importance
Signal Processing	PyWavelets	1.5+	CWT for acoustic signals
Imbalanced Data	imbalanced-learn	0.11+	SMOTE oversampling
Edge Export	ONNX + onnxruntime	1.16+	Cross-platform inference
Optimization	TensorRT (Nvidia)	8.6+	INT8 edge quantization
API Server	FastAPI + Uvicorn	0.104+	Backend REST + WebSocket
Data	Pandas + NumPy	2.0+ / 1.26+	Time-series wrangling
Frontend	Node.js	20+	Next.js runtime
Framework	Next.js	14+	App Router SSR/SSG
3D	React Three Fiber + Drei	8+ / 9+	WebGL 3D aircraft
Charts	Recharts	2.10+	Live sensor streaming
Styling	Tailwind CSS	3.4+	Dark theme UI
State	Zustand	4.5+	Real-time sensor state
Probabilistic ML	pgmpy	0.1.25+	Bayesian network (ADAPT)
NLP	spaCy	3.7+	Text preprocessing
Gradient Boosting	XGBoost + LightGBM	2.0+ / 4.0+	Landing gear, APU faults

7.2 Compute Requirements

Resource	Minimum	Recommended	Notes
GPU (training)	NVIDIA RTX 3060 (8GB VRAM)	RTX 4090 / A100 (24GB)	Required for LLaMA/Gemma fine-tuning
CPU	8-core modern CPU	16-core (AMD Ryzen 9 / Intel i9)	Parallel dataset preprocessing
RAM	16 GB	32–64 GB	AeroTwin 15.1M point dataset
Storage	50 GB SSD	500 GB NVMe SSD	All datasets + model checkpoints
Edge inference	Any modern CPU	NVIDIA Jetson Orin	ONNX runtime runs on CPU/GPU
Cloud (demo)	Vercel hobby (free)	Railway Pro + Vercel Pro	FastAPI + Next.js deployment

7.3 Data Download Checklist

✓	Dataset	Source URL / Repository
◻	NASA C-MAPSS (FD001–FD004)	nasa.gov/prognostics-center-of-excellence

✓	Dataset	Source URL / Repository
■	UCI Hydraulic Systems	archive.ics.uci.edu/ml/datasets/Condition+monitoring+of+hydraulic+systems
■	NASA ADAPT Electrical	ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository
■	NASA Li-ion Aging + HIRF	ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository
■	NASA Ames CFRP Composites	nasa.gov/prognostics-center-of-excellence
■	NGAFID-MC QAR Benchmark	ngafid.org
■	MaintNet Aviation Corpus	maintnet.org
■	AeroTwin Boeing 787 LG	Research paper dataset (see Section 3)
■	LLaMA-3.2-3B weights	huggingface.co/meta-llama/Llama-3.2-3B
■	Gemma-3-4B weights	huggingface.co/google/gemma-3-4b

8. HACKATHON IMPLEMENTATION ROADMAP

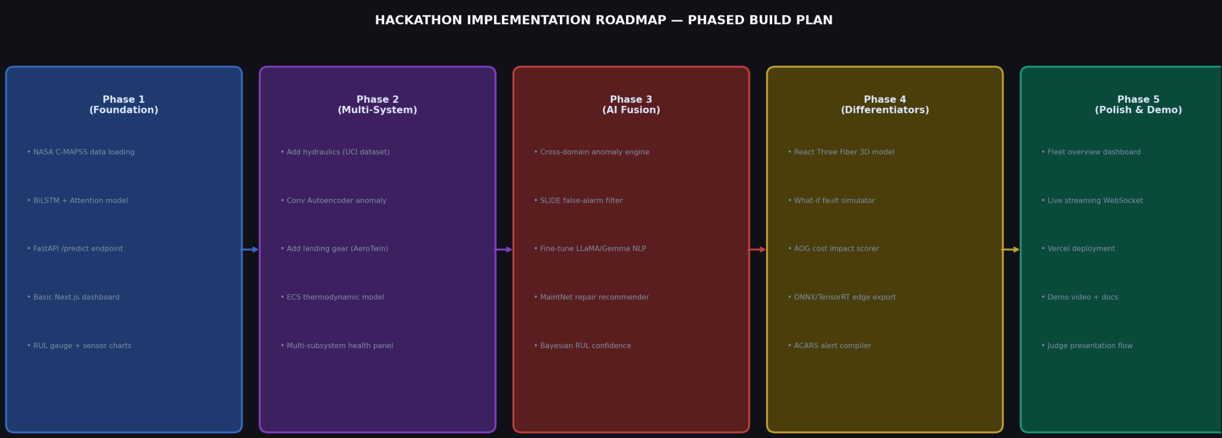

Figure 4: 5-Phase Hackathon Build Roadmap — sequenced for maximum demo impact

Phase 1 — Foundation (Hours 1–8)

- Load and preprocess NASA C-MAPSS (FD001, FD004) — normalize, build sliding windows
- Train BiLSTM+Attention model (PyTorch) — target RMSE < 15 cycles on test set
- SHAP feature selection — reduce 21 → 14 sensors, generate importance bar chart
- FastAPI /predict endpoint with Pydantic model — returns RUL estimate + confidence
- Basic Next.js dashboard — engine health card + RUL countdown gauge

Phase 2 — Multi-System Expansion (Hours 8–20)

- Load UCI Hydraulic dataset — train 1D Conv Autoencoder on nominal sequences
- Load AeroTwin dataset — XGBoost classifier for landing gear bushing wear multi-class
- Boeing 737-200 ECS thermodynamic model — delta-T anomaly scoring per sensor point
- Extend FastAPI to /predict/{subsystem} routing — hydraulics, landing gear, ECS endpoints
- Multi-panel dashboard — subsystem health grid with traffic-light status indicators

Phase 3 — Cross-Domain Fusion (Hours 20–32)

- Implement cross-domain anomaly engine — ECS→engine bleed correlation scoring
- SLIDE hierarchical fault tree — suppress false alarms in tightly-coupled systems
- Fine-tune LLaMA-3.2-3B on MaintNet corpus — repair step recommendation API
- TF-IDF + SVD clustering — auto-categorize maintenance logs into fault topics
- ACARS alert compiler — generate ≤220-char compressed diagnostic messages

Phase 4 — Differentiators (Hours 32–44)

- React Three Fiber 3D aircraft model — component health heatmap overlays
- What-if fault injection slider — real-time RUL cascade updates on fault injection
- AOG cost impact panel — probabilistic financial risk calculator per component
- LLM repair chat interface — maintenance engineer Q&A; powered by fine-tuned model
- ONNX export + INT8 quantization — sub-5ms latency benchmark panel

Phase 5 — Polish & Demo (Hours 44–48)

- Fleet maintenance Gantt planner — multi-aircraft consolidated scheduling view
- WebSocket live streaming — real-time sensor data animation in all chart panels
- Vercel deployment — shareable demo URL for judges
- Demo flow rehearsal — 5-minute structured presentation: problem → architecture → live demo → impact

- Documentation — README, architecture diagram, dataset citations, API docs

9. REFERENCE PAPERS & CITATIONS

Engine / Turbofan PHM

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10. EXECUTIVE SUMMARY

What you are building: A holistic, fleet-wide Aircraft Predictive Health Management system that monitors 12 aircraft subsystems simultaneously using 9+ real-world research datasets, an ensemble of 17+ specialized ML models, and a Next.js dashboard featuring a clickable 3D aircraft model with live sensor streaming.

How it beats the competition: While nearly every other team will train a single LSTM on C-MAPSS and plot a RUL curve, your system demonstrates cross-domain fault cascade analysis (ECS-engine coupling, hydraulic-actuator interactions), edge deployment with quantified <5ms latency, LLM-powered repair recommendations from real maintenance logs, an interactive what-if fault simulator, and quantified AOG business impact scoring.

The key differentiators in order of judge impact:

Rank	Differentiator	Why Judges Will Remember It
1	3D Aircraft Inspector with fault heatmaps	No competing team will show a clickable 3D aircraft with live health data
2	What-if fault injection simulator	Live interactive demo — judges can inject faults and watch RUL collapse
3	Cross-domain fault cascade graph	Proves architectural sophistication beyond single-component analysis
4	LLM repair recommendation chat	Bridges ML output to human actionable steps — NLP+PHM integration
5	AOG $150k/hr business impact panel	Quantified financial value — directly relevant to aviation operators
6	ACARS 220-char alert compiler	Demonstrates real engineering constraint awareness for aviation deployment
7	Fleet maintenance Gantt planner	System-level operational thinking, not just single-component ML

Good luck at the hackathon. Build the 3D visual first — it is your first impression and your last impression.